

Fuzzy-MambaHSI: A Deep Neuro-Fuzzy State Space Model for Hyperspectral Image Classification

1. Motivation and Core Idea

Hyperspectral image (HSI) classification faces two fundamental challenges that existing methods address only partially:

Challenge 1 — Spectral Ambiguity. The spectral signatures of different land-cover materials frequently overlap, and mixed pixels (containing multiple materials within a single pixel footprint) introduce inherent uncertainty into the feature space. This uncertainty is *fuzzy* in nature: a pixel’s membership in a given class is often a matter of degree, not a binary decision.

Challenge 2 — Long-Range Spatial Dependence at Linear Cost. Pixel-level classification demands that every pixel’s representation encode contextual information from distant regions. Transformers achieve this but at quadratic cost; CNNs are limited to local receptive fields. Mamba (selective state space models) offers linear-complexity long-range modeling, but its state transitions are purely data-driven and lack an explicit mechanism for handling the fuzzy nature of spectral data.

Our proposal — Fuzzy-MambaHSI bridges these two challenges by embedding a TSK-style fuzzy inference system directly into the Mamba-based architecture. Rather than applying fuzzy logic as a post-hoc adapter (as in Fuzzy-ViT’s cross-domain setting), we integrate it *within* the state space dynamics, the spectral grouping mechanism, and the spatial-spectral fusion stage, producing an architecture whose inductive biases are natively aligned with the spectral uncertainty inherent in HSI data.

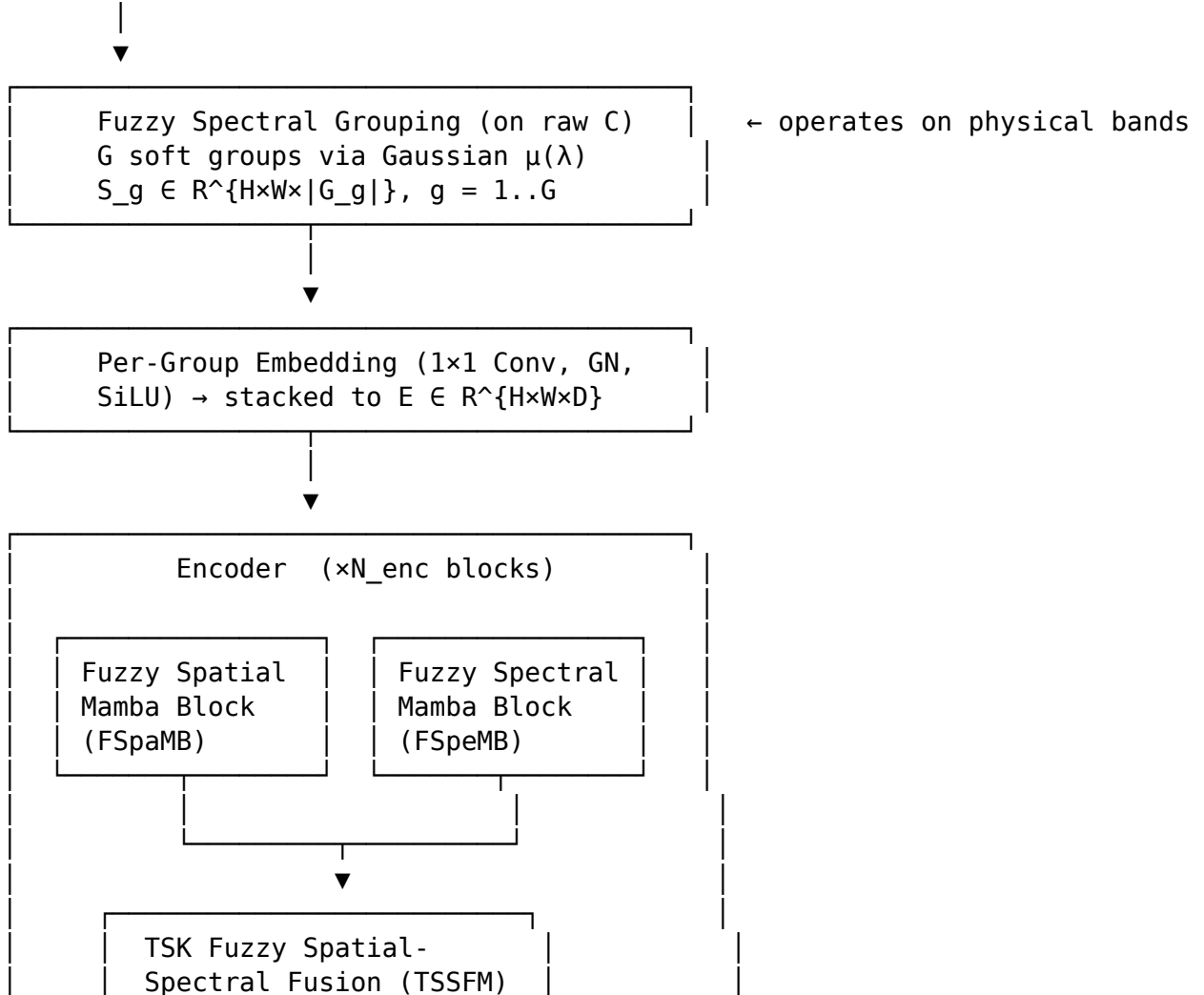
Distinction from Prior Work

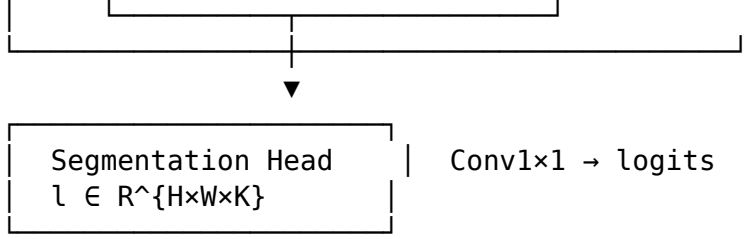
Aspect	MambaHSI	Fuzzy-ViT	Fuzzy-MambaHSI (Ours)
Backbone	Mamba (SSM)	ViT (Attention)	Mamba (SSM)
Fuzzy integration	None	Sigmoid membership + learnable rules as cross-domain adapter	Gaussian TSK membership embedded in Δ -modulation, raw-band spectral grouping, and pixel-wise fusion
Spectral handling	Hard group split on embedding channels + Mamba	N/A (RGB/medical)	Fuzzy membership grouping on raw bands + per-group embedding + Mamba

Aspect	MambaHSI	Fuzzy-ViT	Fuzzy-MambaHSI (Ours)
Spatial-spectral fusion	Two global scalar weights	N/A	Pixel-wise first-order TSK fuzzy inference
Complexity	$O(L)$	$O(L^2)$ for attention	$O(L)$ preserved (see §6 for constants)
Primary setting	HSI classification	Cross-domain transfer (general → medical)	HSI classification with explicit uncertainty modeling

2. Overall Architecture

Input HSI $I \in \mathbb{R}^{H \times W \times C}$





Three strategic points of fuzzy integration are preserved from v1, but the **first** point has moved *upstream of embedding*, not inside it:

1. **Fuzzy Spectral Grouping + FSpaMB** — soft band-level partitioning, followed by Mamba modeling of inter-group relations.
2. **FSpaMB** — fuzzy-modulated spatial state transitions (Δ -modulation).
3. **TSSFm** — pixel-wise, content-dependent TSK fusion replacing global scalar weights.

3. Module Design

3.1 Fuzzy Spatial Mamba Block (FSpaMB)

3.1.1 Role In standard Mamba, the selectivity mechanism makes **B**, **C**, and Δ input-dependent via linear projections. FSpaMB *enriches* this selectivity with a fuzzy-rule-based modulation of Δ : multiple rules compete to set the effective time-scale per token, yielding multi-modal state dynamics. We position this as a **functional** improvement — enabling mixture-of-experts-like behavior over the state dynamics — and reserve the interpretability question for empirical validation (§7).

3.1.2 Fuzzy Δ -Modulation Define a learnable fuzzy rule base over the embedding space:

$$\mathbf{m}_r^{\text{spa}} \in \mathbb{R}^{D'}, \quad \sigma_r^{\text{spa}} \in \mathbb{R}_{>0}^{D'}, \quad \delta_r \in \mathbb{R}^D, \quad r = 1, \dots, R_{\text{spa}}$$

where $D' \ll D$ is a low-rank projection dimension used only for rule activation (keeping rule evaluation cheap). Given a flattened spatial token $\mathbf{x}_t \in \mathbb{R}^D$:

$$\tilde{\mathbf{x}}_t = \mathbf{W}_{\text{proj}} \mathbf{x}_t \in \mathbb{R}^{D'} \tag{1}$$

$$\mu_{r,d}^{\text{spa}}(\tilde{x}_{t,d}) = \exp\left(-\frac{(\tilde{x}_{t,d} - m_{r,d}^{\text{spa}})^2}{2(\sigma_{r,d}^{\text{spa}})^2}\right) \tag{2}$$

$$f_r^{\text{spa}}(\mathbf{x}_t) = \prod_{d=1}^{D'} \mu_{r,d}^{\text{spa}}(\tilde{x}_{t,d}), \quad \bar{f}_r^{\text{spa}}(\mathbf{x}_t) = \frac{f_r^{\text{spa}}(\mathbf{x}_t)}{\sum_{i=1}^{R_{\text{spa}}} f_i^{\text{spa}}(\mathbf{x}_t)} \tag{3}$$

The fuzzy Δ -offset is then the TSK-weighted consequent:

$$\Delta_t^{\text{fuzzy}} = \sum_{r=1}^{R_{\text{spa}}} \bar{f}_r^{\text{spa}}(\mathbf{x}_t) \cdot \delta_r \quad (4)$$

$$\Delta_t = \text{softplus}(\mathbf{W}_{\Delta} \mathbf{x}_t + \lambda(t) \cdot \Delta_t^{\text{fuzzy}}) \quad (5)$$

where $\lambda(t) \in [0, 1]$ is the warm-up coefficient (§5.1) that gates the fuzzy contribution during early training. Discretized state matrices $\bar{\mathbf{A}}_t, \bar{\mathbf{B}}_t$ are computed from Δ_t via zero-order hold and the recurrence proceeds as standard Mamba:

$$\mathbf{h}_t = \bar{\mathbf{A}}_t \mathbf{h}_{t-1} + \bar{\mathbf{B}}_t \mathbf{x}_t, \quad y_t = \mathbf{C}_t \mathbf{h}_t \quad (6)$$

3.1.3 FSpaMB Forward Pass

$$\begin{aligned} \mathbf{HF}_{\text{spa}} &= \text{Flatten}(\mathbf{H}^i) \in \mathbb{R}^{B \times L_1 \times D} \\ \mathbf{HR}_{\text{spa}} &= \text{SiLU}\left(\text{GN}\left(\text{FuzzyMamba}_{\text{spa}}(\mathbf{HF}_{\text{spa}})\right)\right) \\ \mathbf{H}_{\text{spa}}^o &= \text{Reshape}(\mathbf{HR}_{\text{spa}}) + \mathbf{H}^i \end{aligned} \quad (7)$$

3.1.4 Honest Statement on Interpretability The rules $\{r = 1, \dots, R_{\text{spa}}\}$ partition the embedding space into soft regions, each with its own state-dynamics preference δ_r . Whether these regions align with semantically meaningful “spatial context modes” (e.g., vegetation vs. urban textures) is an *empirical* question. We commit to the following post-hoc analyses in §7 before any interpretability claim is made: (a) visualizing per-rule activation maps $\bar{f}_r^{\text{spa}}(\mathbf{x}_t)$ over test images; (b) reporting per-class mean activation $\mathbb{E}[\bar{f}_r^{\text{spa}} \mid y = k]$ to check whether rules specialize to land-cover classes; (c) measuring rule-usage entropy to detect rule collapse. Until these analyses are reported, FSpaMB is presented purely as a multi-modal Δ -modulation mechanism.

3.2 Fuzzy Spectral Mamba Block (FSpaMB) — *Restructured*

3.2.1 Why the Redesign In v1, fuzzy spectral grouping was defined on the embedding dimension index $d \in \{1, \dots, D\}$. After 1×1 convolution, however, each channel d is already a learned mixture of original bands, so Gaussian centers over d do not capture “spectral locality” in any physical sense. In v2 we move fuzzy grouping **upstream of embedding**, directly onto the raw spectral axis $\lambda \in \{1, \dots, C\}$, where the index is wavelength-ordered and physically meaningful.

3.2.2 Fuzzy Spectral Grouping on Raw Bands Define G learnable group centers and widths in band-index space:

$$c_g \in [1, C], \quad \sigma_g \in \mathbb{R}_{>0}, \quad g = 1, \dots, G \quad (8)$$

(Initialization: c_g evenly spaced over $[1, C]$; σ_g initialized so neighboring group supports overlap by $\approx 30\%$ at half-maximum.)

For each raw band λ , compute its soft membership to each group:

$$\alpha_{g,\lambda} = \frac{\exp\left(-\frac{(\lambda-c_g)^2}{2\sigma_g^2}\right)}{\sum_{j=1}^G \exp\left(-\frac{(\lambda-c_j)^2}{2\sigma_j^2}\right)} \quad (9)$$

Each group then constructs a soft band-weighted HSI slice:

$$\mathbf{S}_g[i, j, \lambda] = \alpha_{g,\lambda} \cdot \mathbf{I}[i, j, \lambda], \quad \lambda = 1, \dots, C \quad (10)$$

Because $\alpha_{g,\lambda}$ is concentrated near c_g , $\mathbf{S}_g$ effectively retains the subset of bands that group g “owns,” but with soft (differentiable) boundaries that let adjacent bands participate in multiple groups — reflecting the continuous nature of spectral absorption features.

3.2.3 Per-Group Embedding Each group-specific HSI slice is embedded independently:

$$\mathbf{E}_g = \text{SiLU}\left(\text{GN}\left(\text{Conv}_{1 \times 1}^{(g)}(\mathbf{S}_g)\right)\right) \in \mathbb{R}^{H \times W \times M} \quad (11)$$

where $M = D/G$. Stacking groups along a new “group” axis gives the embedding used by the encoder:

$$\mathbf{E} \in \mathbb{R}^{H \times W \times G \times M}, \quad \text{equivalent to } \mathbb{R}^{H \times W \times D} \text{ after concatenation} \quad (12)$$

This is the embedding that enters FSpaMB. The crucial distinction from MambaHSI is that the block structure $G \times M$ is now **semantically grounded in spectral band groups**, not arbitrary channel partitioning.

3.2.4 FSpaMB: Inter-Group Mamba with TSK Defuzzification Within the encoder, FSpaMB operates on the group axis. Let $\mathbf{H}^i \in \mathbb{R}^{B \times H \times W \times G \times M}$ be the input with group structure preserved:

$$\begin{aligned} \mathbf{H}\mathbf{F}_{\text{spe}} &= \text{Flatten}(\mathbf{H}^i) \in \mathbb{R}^{N \times G \times M}, \quad N = B \cdot H \cdot W \\ \mathbf{H}\mathbf{R}_{\text{spe}} &= \text{SiLU}\left(\text{GN}\left(\text{Mamba}(\mathbf{H}\mathbf{F}_{\text{spe}})\right)\right) \in \mathbb{R}^{N \times G \times M} \end{aligned} \quad (13)$$

The Mamba here models inter-group sequence relations, as in MambaHSI. For defuzzification, we use the **group firing strength** derived from the mean band membership:

$$\phi_g = \frac{1}{C} \sum_{\lambda=1}^C \alpha_{g,\lambda}, \quad \bar{\phi}_g = \frac{\phi_g}{\sum_j \phi_j} \quad (14)$$

The TSK output fuses groups via a first-order consequent:

$$\mathbf{H}_{\text{spe}}^o[i, j] = \mathbf{H}^i[i, j] + \sum_{g=1}^G \bar{\phi}_g (\mathbf{W}_g^{\text{up}} \mathbf{H} \mathbf{R}_{\text{spe}}[i, j, g, :] + \mathbf{b}_g) \in \mathbb{R}^D \quad (15)$$

3.2.5 Summary of the Redesign The restructured FSpeMB now has a **genuine** physical interpretation: c_g is a learned central wavelength for group g , σ_g is its bandwidth, and ϕ_g is the group's overall spectral coverage weight. This directly addresses the reviewer's concern that v1 was computing "latent-channel fuzzy grouping" disguised as spectral grouping.

3.3 TSK Fuzzy Spatial-Spectral Fusion Module (TSSFM)

This module is retained from v1 with only minor refinements. It remains the innovation most likely to deliver measurable gains, because it directly replaces the coarse two-scalar fusion of MambaHSI with pixel-wise content-dependent blending.

3.3.1 Formulation Define R_f fusion rules. Compute a compact antecedent representation:

$$\mathbf{z}_{ij} = \mathbf{W}_z [\mathbf{H}_{\text{spa}}^o[i, j] \parallel \mathbf{H}_{\text{spe}}^o[i, j]] + \mathbf{b}_z \in \mathbb{R}^{D_z} \quad (16)$$

Rule activations (Gaussian antecedent):

$$f_r^{\text{fus}}(\mathbf{z}_{ij}) = \prod_{d=1}^{D_z} \exp\left(-\frac{(z_{ij,d} - m_{r,d}^{\text{fus}})^2}{2(\sigma_{r,d}^{\text{fus}})^2}\right), \quad \bar{f}_r^{\text{fus}}(\mathbf{z}_{ij}) = \frac{f_r^{\text{fus}}(\mathbf{z}_{ij})}{\sum_i f_i^{\text{fus}}(\mathbf{z}_{ij})} \quad (17)$$

First-order TSK consequent (vector-valued, rather than scalar weighting):

$$\hat{\mathbf{y}}_r(\mathbf{z}_{ij}) = \mathbf{W}_r^{\text{fus}} \mathbf{z}_{ij} + \mathbf{b}_r^{\text{fus}} \in \mathbb{R}^D \quad (18)$$

Defuzzified, residual-connected fused output:

$$\mathbf{H}_{\text{fus}}[i, j] = \mathbf{H}^i[i, j] + \sum_{r=1}^{R_f} \bar{f}_r^{\text{fus}}(\mathbf{z}_{ij}) \cdot \hat{\mathbf{y}}_r(\mathbf{z}_{ij}) \quad (19)$$

The consequents are linear functions, so each rule simultaneously *blends* and *transforms* features — strictly more expressive than any scalar-weighted sum of $\mathbf{H}_{\text{spa}}^o$ and $\mathbf{H}_{\text{spe}}^o$. Crucially, $\mathbf{z}_{ij}$ is computed **per pixel**, so the fusion adapts locally: a boundary pixel can rely on spatial features while an interior vegetation pixel can rely on spectral features.

4. Theoretical Justification

4.1 Why Gaussian TSK Rather than Sigmoid Membership

Fuzzy-ViT uses $M = \sigma(XW_r^T)$, a sigmoid membership which acts as a soft cross-domain gate. Gaussian membership is better suited to HSI for three reasons:

1. **Physical locality.** With the v2 redesign, spectral group centers c_g live in band-index space. Gaussian membership directly encodes the localized absorption structure of real spectra; a sigmoid projection would instead compute a global correlation that discards band ordering.
2. **TSK universal approximation.** A first-order TSK system with Gaussian antecedents and linear consequents is a universal approximator (Wang, 1992). The full TSSFm pipeline (antecedent $\rightarrow$ activation $\rightarrow$ linear consequent $\rightarrow$ weighted defuzzification) inherits this property, which the simpler sigmoid formulation does not.
3. **Post-hoc diagnosability.** Centers m_r and widths σ_r are inspectable scalars — we can plot them, cluster them, and check whether they stabilize during training. This enables the validation protocol in §7.

4.2 Expressiveness as a Hierarchical Fuzzy System

The three modules form a three-level fuzzy reasoning chain: - **Level 1 (FSpeMB, upstream of embedding):** Fuzzy rules determine *what is grouped* — the composition of spectral features. - **Level 2 (FSpaMB):** Fuzzy rules determine *how context flows* — the dynamics of the state space. - **Level 3 (TSSFm):** Fuzzy rules determine *how sources are combined* — the fusion of spatial and spectral representations.

Each level addresses a distinct source of uncertainty (composition, dynamics, fusion), and their composition is end-to-end differentiable.

5. Training Strategy

5.1 Staged / Warm-Up Training

Randomly initialized Gaussian parameters can saturate or collapse if trained alongside the full network from step zero. We adopt a warm-up schedule:

$$\lambda(t) = \min\left(1, \frac{t}{T_{\text{warm}}}\right), \quad t = 1, 2, \dots \quad (20)$$

$\lambda(t)$ multiplies the fuzzy contribution inside FSpaMB (Eq. 5) and — optionally — inside TSSFM. For the first T_{warm} epochs the network behaves close to standard MambaHSI (modulo fuzzy spectral grouping, which is active from the start because it replaces the embedding itself).

5.2 Loss Function

Classification loss follows MambaHSI:

$$\mathcal{L}_{\text{cls}} = \text{CrossEntropy}(\mathbf{1}, \mathbf{Y}_{\text{tr}}) \quad (21)$$

5.3 Rule Diversity Regularization (Revised)

The v1 regularizer $\mathcal{L}_{\text{div}} = -\frac{1}{R(R-1)} \sum_{r \neq s} \|\mathbf{m}_r - \mathbf{m}_s\|_2$ unconditionally rewards pushing centers apart, which can drive them outside the data distribution. We replace it with a hinge-style separation penalty that only activates when centers are too close:

$$\mathcal{L}_{\text{div}} = \frac{1}{R(R-1)} \sum_{r \neq s} \max\left(0, \tau - \frac{\|\mathbf{m}_r - \mathbf{m}_s\|_2}{\|\mathbf{m}_r\|_2 + \|\mathbf{m}_s\|_2 + \epsilon}\right)^2 \quad (22)$$

Here the separation is **normalized by the centers' own scales**, so the penalty does not grow unbounded as $\|\mathbf{m}\|_2$ increases. The threshold τ (e.g., 0.1) defines a minimum relative spacing below which duplicate rules are penalized. Above τ , the gradient is zero — centers are free to settle wherever the data prefer.

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \beta \cdot \mathcal{L}_{\text{div}} \quad (23)$$

with $\beta \approx 0.01$.